

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 14: Corner Detection

WORKSHEET SOLUTIONS

1. 1. What is a corner in an image?

A corner is a point in an image where two or more edges meet and the intensity changes significantly in all 2D directions (both horizontal and vertical).

2. 2. Name one corner detection algorithm.

Harris Corner Detector (or Shi-Tomasi Corner Detector / FAST Corner Detector).

3. 3. Why are corners useful in object tracking?

Corners are highly distinctive, stable, and invariant to rotation and illumination changes. Unlike flat regions or edges (which suffer from the aperture problem), corners provide unique 2D localized spatial information that can be precisely matched across consecutive video frames.

4. 4. Difference between edge and corner.

Feature	Edge	Corner
Intensity Variation	Significant change in one direction (perpendicular to the edge).	Significant change in all directions (both 2D axes).
Localization	Poor 2D localization (suffers from aperture problem along the edge line).	High 2D localization (uniquely identifies a single point).